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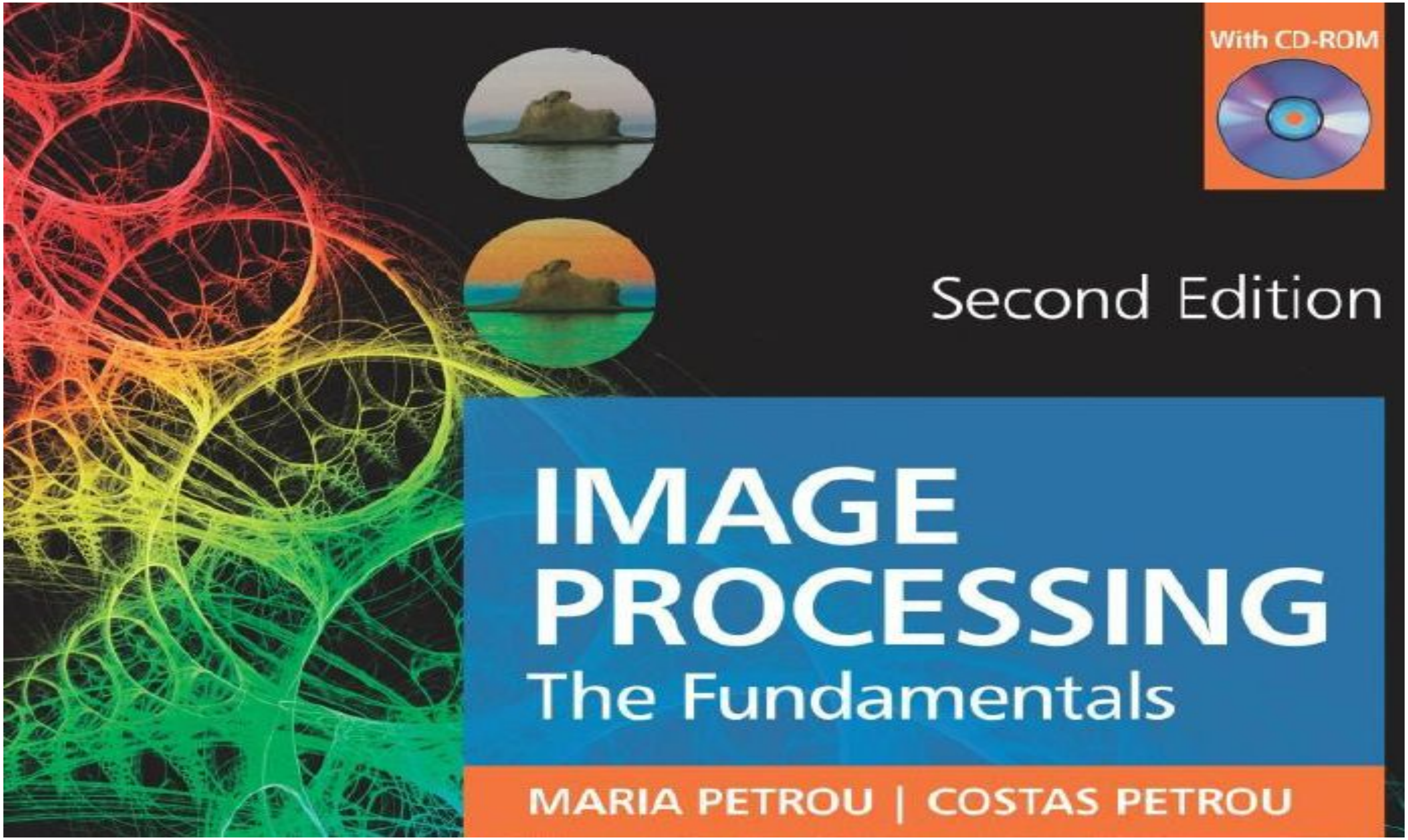
Image Processing Part of CIT-421



Chapter 1 : Introduction

Why do we process images?

- Image Processing has been developed in response to three major problems concerned with pictures:

- *Picture digitization and coding to facilitate transmission, printing and storage of pictures;*

- *Picture enhancement and restoration in order, for example, to interpret more easily pictures of the surface of other planets taken by various probes;*

- *picture segmentation and description as an early stage to Machine Vision.*

- *Image Processing nowadays refers mainly to the processing of digital images.*

What is an image?

- A **panchromatic image** is a 2D light intensity function, $f(x, y)$, where x and y are spatial coordinates and the value of f at (x, y) is proportional to the brightness of the scene at that point.
- If we have a **multispectral image**, $f(x, y)$ is a vector, each component of which indicates the brightness of the scene at point (x, y) at the corresponding spectral **band**.

What is an image?

- A **panchromatic image:** Panchromatic images appear black and white because the band is a combination of RGB that are usually collected separately. This allows for greater spatial resolution at the expense of being able to differentiate color.

- Information contained in each pixel of a panchromatic image, which is reflected by the objects in the pixel and is detected by the satellite sensor.

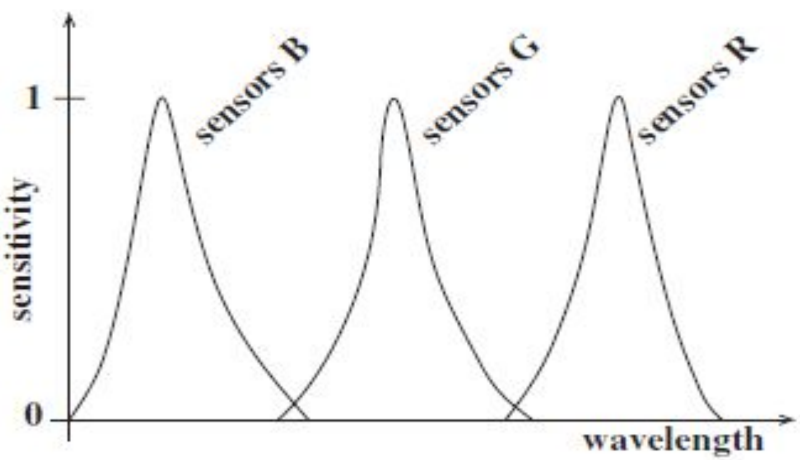
What is a digital image?

- A **digital image** is an image $f(x, y)$ that has been discretized both in spatial coordinates and in brightness. It is represented by a 2D integer array, or a series of 2D arrays, one for each color band. The digitized brightness value is called **grey level**.
- Each element of the array is called **pixel** or **pel**, derived from the term "picture element".
- Usually, the size of such an array is a few hundred pixels by a few hundred pixels and there are several dozens of possible different grey levels. Thus, a digital image looks like this with $0 \leq f(x, y) \leq G-1$, where usually N and G are expressed as positive integer powers of $2\ (N = 2^n, G = 2^m)$.

$$f(x,y) = \begin{bmatrix} f(1,1) & f(1,2) & \dots & f(1,N) \\ f(2,1) & f(2,2) & \dots & f(2,N) \\ \vdots & \vdots & & \vdots \\ f(N,1) & f(N,2) & \dots & f(N,N) \end{bmatrix} \tag{1.1}$$

What is a spectral band?

- A color band is a range of wavelengths of the electromagnetic spectrum, over which the sensors we use to capture an image have nonzero sensitivity.
- Typical color images consist of three color bands. This means that they have been captured by three different sets of sensors, each set made to have a different sensitivity function.
- Figure 1.1 shows typical sensitivity curves of a multispectral camera.



- This recorded value is the brightness of the image in the location of the sensor and in the band of the sensor. This figure shows the sensitivity curves of three different sensor types.

Why do most image processing algorithms refer to grey images, while most images we come across are colour images?

For various reasons.

1) A lot of the processes we apply to a grey image can be easily extended to a colour image by applying them to each band separately.

2) A lot of the information conveyed by an image is expressed in its grey form and so colour is not necessary for its extraction. That is the reason black and white television receivers had been perfectly acceptable to the public for many years and black and white photography is still popular with many photographers.

3) For many years colour digital cameras were expensive and not widely available. A lot of image processing techniques were developed around the type of image that was available. These techniques have been well established in image processing.

How is a digital image formed?

- Each pixel of an image corresponds to a part of a physical object in the 3D world. This physical object is illuminated by some light which is partly reflected and partly absorbed by it.
- Part of the reflected light reaches the array of sensors used to image the scene and is responsible for the values recorded by these sensors.
- One of these sensors makes up a pixel and its field of view corresponds to a small patch in the imaged scene. The recorded value by each sensor depends on its sensitivity curve.

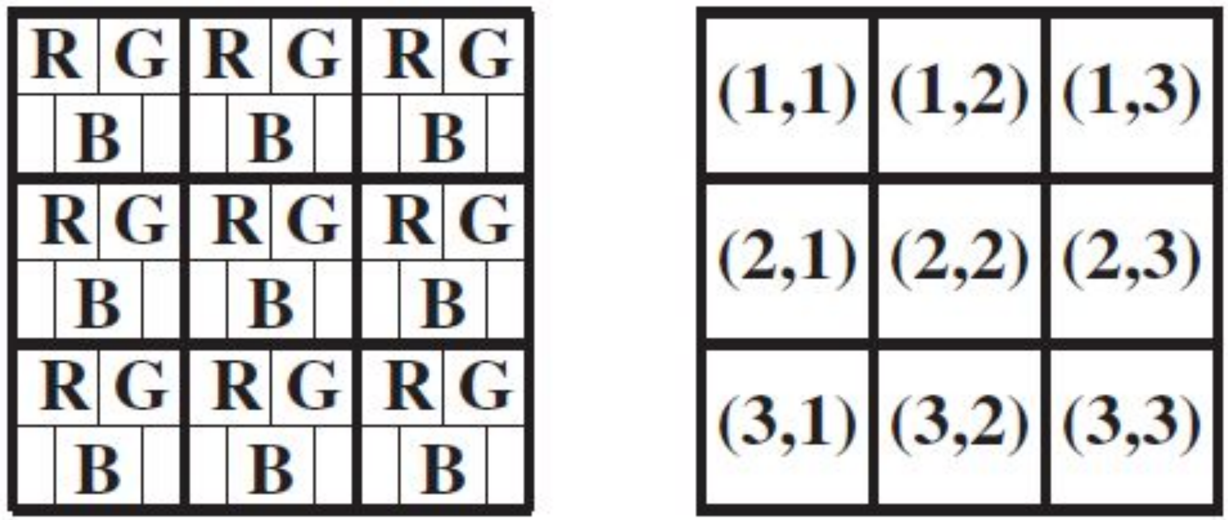
- When a photon of a certain wavelength reaches the sensor, its energy is multiplied with the value of the sensitivity curve of the sensor at that wavelength and is accumulated.
- ***The total energy collected by the sensor (during the exposure time) is eventually used to compute the grey value of the pixel that corresponds to this sensor.***

- **Example 1.1: You are given a triple array of 3×3 sensors, arranged as shown in figure 1.3, with their sampled sensitivity curves given in table 1.1. The last column of this table gives in some arbitrary units the energy, $E\lambda i$, carried by photons with wavelength λi.**

Wavelength	Sensors B	Sensors G	Sensors R	Energy
λ_0	0.2	0.0	0.0	1.00
λ_1	0.4	0.2	0.1	0.95
λ_2	0.8	0.3	0.2	0.90
λ_3	1.0	0.4	0.2	0.88
λ_4	0.7	0.6	0.3	0.85
λ_5	0.2	1.0	0.5	0.81
λ_6	0.1	0.8	0.6	0.78
λ_7	0.0	0.6	0.8	0.70
λ_8	0.0	0.3	1.0	0.60
λ_9	0.0	0.0	0.6	0.50

- **Table 1.1: The sensitivity curves of three types of sensor for wavelengths in the range [$\lambda 0$, $\lambda 9$], and the corresponding energy of photons of these particular wavelengths in some arbitrary units.**

- **The shutter of the camera is open long enough for 10 photons to reach the locations of the sensors.**



- **Figure 1.3: Three** 3×3 **sensor arrays interlaced. On the right, the locations of the pixels they make up. Although the three types of sensor are slightly misplaced with respect to each other, we assume that each triplet is coincident and forms a single pixel.**

- **The wavelengths of the photons that reach the pixel locations of each triple sensor, as identified in figure 1.3, are:**

- **Location** (1, 1)**:** $\lambda 0,\lambda 9,\lambda 9,\lambda 8,\lambda 7,\lambda 8,\lambda 1,\lambda 0,\lambda 1,\lambda 1$
- **Location** (1, 2)**:** $\lambda 1,\lambda 3,\lambda 3,\lambda 4,\lambda 4,\lambda 5,\lambda 2,\lambda 6,\lambda 4,\lambda 5$
- **Location** (1, 3)**:** $\lambda 6,\lambda 7,\lambda 7,\lambda 0,\lambda 5,\lambda 6,\lambda 6,\lambda 1,\lambda 5,\lambda 9$
- **Location** (2, 1)**:** $\lambda 0,\lambda 1,\lambda 0,\lambda 2,\lambda 1,\lambda 1,\lambda 4,\lambda 3,\lambda 3,\lambda 1$
- **Location** (2, 2)**:** $\lambda 3,\lambda 3,\lambda 4,\lambda 3,\lambda 4,\lambda 4,\lambda 5,\lambda 2,\lambda 9,\lambda 4$
- **Location** (2, 3)**:** $\lambda 7,\lambda 7,\lambda 6,\lambda 7,\lambda 6,\lambda 1,\lambda 5,\lambda 9,\lambda 8,\lambda 7$
- **Location** (3, 1)**:** $\lambda 6,\lambda 6,\lambda 1,\lambda 8,\lambda 7,\lambda 8,\lambda 9,\lambda 9,\lambda 8,\lambda 7$
- **Location** (3, 2)**:** $\lambda 0,\lambda 4,\lambda 3,\lambda 4,\lambda 1,\lambda 5,\lambda 4,\lambda 0,\lambda 2,\lambda 1$
- **Location** (3, 3)**:** $\lambda 3,\lambda 4,\lambda 1,\lambda 0,\lambda 0,\lambda 4,\lambda 2,\lambda 5,\lambda 2,\lambda 4$

- **Calculate the values that each sensor array will record and thus produce the three** photon energy **bands recorded.**

- *We denote by gX*(*i, j*) *the value that will be recorded by sensor type X at location* (*i, j*). *For sensors R, G and B in location* (1, 1)*, the recorded values will be:*

$$\begin{aligned}
g_R(1,1) &= 2E_{\lambda_0} \times 0.0 + 2E_{\lambda_9} \times 0.6 + 2E_{\lambda_8} \times 1.0 + 1E_{\lambda_7} \times 0.8 + 3E_{\lambda_1} \times 0.1 \\
&= 1.0 \times 0.6 + 1.2 \times 1.0 + 0.7 \times 0.8 + 2.85 \times 0.1 \\
&= 2.645 \qquad (1.2) \\
g_G(1,1) &= 2E_{\lambda_0} \times 0.0 + 2E_{\lambda_9} \times 0.0 + 2E_{\lambda_8} \times 0.3 + 1E_{\lambda_7} \times 0.6 + 3E_{\lambda_1} \times 0.2 \\
&= 1.2 \times 0.3 + 0.7 \times 0.6 + 2.85 \times 0.2 \\
&= 1.35 \qquad (1.3) \\
g_B(1,1) &= 2E_{\lambda_0} \times 0.2 + 2E_{\lambda_9} \times 0.0 + 2E_{\lambda_8} \times 0.0 + 1E_{\lambda_7} \times 0.0 + 3E_{\lambda_1} \times 0.4 \\
&= 2.0 \times 0.2 + 2.85 \times 0.4 \\
&= 1.54 \qquad (1.4)
\end{aligned}$$

- *Working in a similar way, we deduce that the energies recorded by the three sensor arrays are:*

$$
\begin{aligned}
E_R &= \begin{pmatrix} 2.645 & 2.670 & 3.729 \\ 1.167 & 4.053 & 4.576 \\ 4.551 & 1.716 & 1.801 \end{pmatrix} \\
E_G &= \begin{pmatrix} 1.350 & 4.938 & 4.522 \\ 2.244 & 4.176 & 4.108 \\ 2.818 & 2.532 & 2.612 \end{pmatrix} \\
E_B &= \begin{pmatrix} 1.540 & 5.047 & 1.138 \\ 4.995 & 5.902 & 0.698 \\ 0.536 & 4.707 & 5.047 \end{pmatrix}
\end{aligned}
\tag{1.5}
$$

What is the physical meaning of the brightness of an image at a pixel position?

- The brightness values of different pixels have been created by using the energies recorded by the corresponding sensors.
- They have significance only *relative* to each other and they are meaningless in absolute terms.
- So, pixel values between different images should only be compared if either care has been taken for the physical processes used to form the two images to be identical, or the brightness values of the two images have somehow been normalized so that the effects of the different physical processes have been removed.
- In that case, we say that the sensors are **calibrated**.

How many bits do we need to store an image?

- The number of bits, b, we need to store an image of size $N \times N$, with 2^m grey levels, is:

$$b = N \times N \times m \qquad (1.6)$$

- So, for a typical 512×512 image with 256 grey levels ($m = 8$) we need 2,097,152 bits or 262,144 8-bit bytes.

- That is why we often try to reduce m and N, without significant loss of image quality.

What determines the quality of an image?

- The quality of an image is a complicated concept, largely subjective and very much application dependent. Basically, an image is of good quality if it is not noisy and

(1) it is not blurred;

(2) it has high resolution;

(3) it has good contrast.

What makes an image blurred?

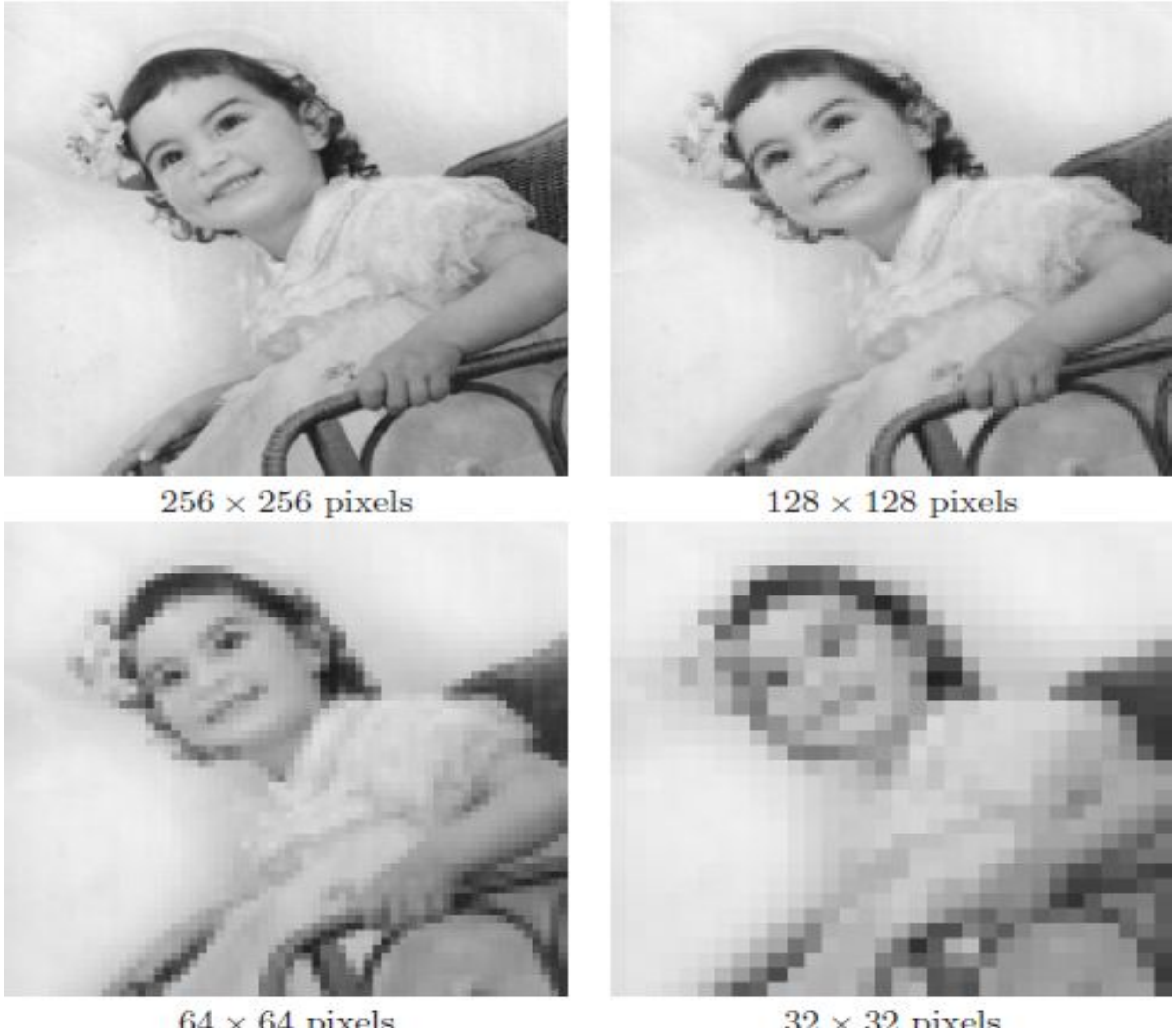
- Image blurring is caused by incorrect image capturing conditions. For example, out of focus camera, or relative motion of the camera and the imaged object.
- The amount of image blurring is expressed by the so called **point spread function** of the imaging system.

What is meant by image resolution?

- Resolution refers to the number of pixels in an image. Resolution is sometimes identified by the width and height of the image as well as the total number of pixels in the image.
- For example, an image that is 2048 pixels wide and 1536 pixels high (2048 x 1536) contains (multiply) 3,145,728 pixels (or 3.1 Megapixels). You could call it a 2048 x 1536 or a 3.1 Megapixel image.
- The resolution of an image expresses how much detail we can see in it and clearly depends on the number of pixels we use to represent a scene (parameter *N* in equation (1.6)) and the number of grey levels used to quantize the brightness values (parameter *m* in equation (1.6))

$$b = N \times N \times m \qquad (1.6)$$

Keeping m constant and decreasing N results in the **checkerboard effect** (figure 1.4).



- Figure 1.4: Keeping the number of grey levels constant and decreasing the number of pixels with which we digitize the same field of view produces the checkerboard effect.

Keeping *N* constant and reducing *m* results in **false contouring** (figure 1.5).

- Figure 1.5: Keeping the size of the image constant (249×199) and reducing the number of grey levels (= $\mathbf{2^m}$) produces false contouring. To display the images, we always map the different grey values to the range [0, 255]

- Experiments have shown that the more detailed a picture is, the less it improves by keeping *N* constant and increasing *m*. So, for a detailed picture, like a picture of crowds (figure 1.6), the number of grey levels we use does not matter much.

- Figure 1.6: Keeping the number of pixels constant and reducing the number of grey levels does not affect much the appearance of an image that contains a lot of details.

256 grey levels ($m = 8$)

128 grey levels ($m = 7$)

64 grey levels ($m = 6$)

32 grey levels ($m = 5$)

16 grey levels ($m = 4$)

8 grey levels ($m = 3$)

4 grey levels ($m = 2$)

2 grey levels ($m = 1$)

What does contrast mean?

- **Contrast** is the difference in luminance or color that makes an object (or its representation in an image or display) distinguishable.
- In visual perception of the real world, contrast is determined by the difference in the color and brightness of the object and other objects within the same field of view.

What does "good contrast" mean?

- Good contrast means that the grey values present in the image range from black to white, making use of the full range of brightness to which the human vision system is sensitive.

What is the purpose of image processing?

Image processing has multiple purposes:

- To improve the quality of an image in a subjective way, usually by increasing its contrast. This is called **image enhancement**.

- To use as few bits as possible to represent the image, with minimum deterioration in its quality. This is called **image compression**.

- To improve an image in an objective way, for example by reducing its blurring. This is called **image restoration**.

- To make explicit certain characteristics of the image which can be used to identify the contents of the image. This is called **feature extraction**.

How do we do image processing?

- We perform image processing by using image transformations. Image transformations are performed using **operators**.
- An operator takes as input an image and produces another
- image.
- There are two kind of operators: **linear operators, and non-linear operators**.

What is the difference between linear and nonlinear operators ?

- Nonlinear operators cannot be collectively characterized. They are usually problem- and application-specific, and they are studied as individual processes used for specific tasks.
- On the contrary, linear operators can be studied collectively, because they share important common characteristics, irrespective of the task they are expected to perform.

What is a linear operator?

- Consider O to be an operator which takes images into images. If f is an image, $O(f)$ is the result of applying O to f. O is linear if

$$O[af + bg] = aO[f] + bO[g] \tag{1.10}$$

- for all images f and g and all scalars a and b.

How are linear operators defined?

- Linear operators are defined in terms of their **point spread functions**. The point spread function of an operator is what we get out if we apply the operator on a point source:

$$O[\text{point source}] \equiv \text{point spread function} \tag{1.11}$$

Or

$$O[\delta(\alpha - x, \beta - y)] \equiv h(x, \alpha, y, \beta) \tag{1.12}$$

- where $\delta(\alpha - x, \beta - y)$ is a point source of brightness 1 centered at point (x, y).

- **What is the relationship between the point spread function of an imaging device and that of a linear operator?**
- They both express the effect of either the imaging device or the operator on a point source.

- Assume that we capture the image of a star by using a camera. The star will appear in the image like a blob: the camera received the light of the point source and spread it into a blob.

- The bigger the blob, the more blurred the image of the star will look.
- So, the point spread function of the camera measures the amount of blurring present in the images captured by this camera. The camera, therefore, acts like a linear operator which accepts as input the ideal brightness function of the continuous real world and produces the recorded digital image.
- That is why we use the term “point spread function” to characterize both cameras and linear operators.

How does a linear operator transform an image?

- If the operator is *linear*, when the point source is *a* times brighter, the result will be *a* times higher:

$$O[a\delta(\alpha - x, \beta - y)] = ah(x, \alpha, y, \beta)$$

(1.13)

- An image is a collection of point sources (the *pixels*) each with its own brightness value.
- For example, assuming that an image *f* is 3 × 3 in size, we may

$$f = \begin{pmatrix} f(1,1) & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & f(1,2) & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} + \cdots + \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & f(3,3) \end{pmatrix} \qquad (1.14)$$

We may say that an image is the sum of these point sources.

How does a linear operator transform an image?

- The effect of an operator characterized by point spread function $h(x, \alpha, y, \beta)$ on an image $f(x, y)$ can be written as:

$$g(\alpha, \beta) = \sum_{x=1}^{N} \sum_{y=1}^{N} f(x, y) h(x, \alpha, y, \beta) \qquad (1.15)$$

- where $g(\alpha, \beta)$ is the output "image", $f(x, y)$ is the input image and the size of the image is $N \times N$.
- Here we treat $f(x, y)$ as the brightness of a point source located at position (x, y).

What is the meaning of the point spread function?

- The point spread function $h(x, \alpha, y, \beta)$ expresses how much the input value at position (x, y) influences the output value at position (α, β).

- If the influence expressed by the point spread function is independent of the actual positions but depends only on the *relative* position of the influencing and the influenced pixels, we have a **shift invariant** ,(if $y(n)$ is the response of the system to $x(n)$, then $y(n-k)$ is the response of the system to $x(n-k)$), point spread function:

$$h(x, \alpha, y, \beta) = h(\alpha - x, \beta - y) \tag{1.16}$$

- Then equation (1.15) is a *convolution*:

$$g(\alpha, \beta) = \sum_{x=1}^{N} \sum_{y=1}^{N} f(x, y) h(\alpha - x, \beta - y) \tag{1.17}$$

- If the columns are influenced independently from the rows of the image, then the point spread function is **separable**:

$$h(x, \alpha, y, \beta) \equiv h_c(x, \alpha) h_r(y, \beta) \tag{1.18}$$

- The above expression serves also as the definition of functions $h_c(x, \alpha)$ and $h_r(y, \beta)$. Then equation (1.15) may be written as a cascade of two 1D transformations:

$$g(\alpha, \beta) = \sum_{x=1}^{N} h_c(x, \alpha) \sum_{y=1}^{N} f(x, y) h_r(y, \beta) \tag{1.19}$$

- If the point spread function is both shift invariant and separable, then equation (1.15) may be written as a cascade of two 1D convolutions:

$$g(\alpha, \beta) = \sum_{x=1}^{N} h_c(\alpha - x) \sum_{y=1}^{N} f(x, y) h_r(\beta - y) \tag{1.20}$$

- **Example 1.5: The following** 3×3 **image**

$$f = \begin{pmatrix} 0 & 2 & 6 \\ 1 & 4 & 7 \\ 3 & 5 & 7 \end{pmatrix} \tag{1.31}$$

- **is processed by a linear operator** O **which has a point spread function** $h(x, \alpha, y, \beta)$ **defined as:**

- $h(1, 1, 1, 1) = 1.0$ $h(1, 1, 1, 2) = 0.5$ $h(1, 1, 1, 3) = 0.0$ $h(1, 2, 1, 1) = 0.5$
- $h(1, 2, 1, 2) = 0.0$ $h(1, 2, 1, 3) = 0.4$ $h(1, 3, 1, 1) = 0.5$ $h(1, 3, 1, 2) = 1.0$
- $h(1, 3, 1, 3) = 0.6$ $h(2, 1, 1, 1) = 0.8$ $h(2, 1, 1, 2) = 0.7$ $h(2, 1, 1, 3) = 0.4$
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- $h(2, 3, 1, 2) = 0.8$ $h(2, 3, 1, 3) = 1.0$ $h(3, 1, 1, 1) = 0.9$ $h(3, 1, 1, 2) = 0.5$
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- $h(1, 1, 3, 3) = 1.0$ $h(1, 2, 3, 1) = 0.5$ $h(1, 2, 3, 2) = 0.1$ $h(1, 2, 3, 3) = 0.6$
- $h(1, 3, 3, 1) = 0.5$ $h(1, 3, 3, 2) = 1.0$ $h(1, 3, 3, 3) = 0.6$ $h(2, 1, 3, 1) = 0.5$
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- $h(3, 1, 3, 1) = 0.8$ $h(3, 1, 3, 2) = 0.5$ $h(3, 1, 3, 3) = 0.5$ $h(3, 2, 3, 1) = 0.5$
- $h(3, 2, 3, 2) = 0.5$ $h(3, 2, 3, 3) = 0.8$ $h(3, 3, 3, 1) = 0.4$ $h(3, 3, 3, 2) = 0.4$
- $h(3, 3, 3, 3) = 1.0$

- **Work out the output image.**
- *The point spread function of the operator h(x, α, y, β) gives the weight with which the pixel value at input position (x, y) contributes to output pixel position (α, β).*
- *Let us call the output image g(α, β). We show next how to calculate g(1, 1). For g(1, 1), we need to use the values of h(x, 1, y, 1) to weigh the values of pixels (x, y) of the input image:*

$$\begin{aligned} g(1,1) &= \sum_{x=1}^{3}\sum_{y=1}^{3} f(x,y)h(x,1,y,1) \\ &= f(1,1)h(1,1,1,1) + f(1,2)h(1,1,2,1) + f(1,3)h(1,1,3,1) \\ &\quad + f(2,1)h(2,1,1,1) + f(2,2)h(2,1,2,1) + f(2,3)h(2,1,3,1) \\ &\quad + f(3,1)h(3,1,1,1) + f(3,2)h(3,1,2,1) + f(3,3)h(3,1,3,1) \\ &= 2\times 1.0 + 6\times 1.0 + 1\times 0.8 + 4\times 0.8 + 7\times 0.5 + 3\times 0.9 \\ &\quad + 5\times 0.7 + 7\times 0.8 = 27.3 \end{aligned} \tag{1.32}$$

- *For* $g(1, 2)$*, we need to use the values of* $h(x, 1, y, 2)$*:*

$$g(1,2) \quad = \quad \sum_{x=1}^{3}\sum_{y=1}^{3} f(x,y)h(x,1,y,2) = 20.1 \tag{1.33}$$

- *The other values are computed in a similar way. Finally, the output image is:*

$$g = \begin{pmatrix} 27.3 & 20.1 & 18.9 \\ 19.4 & 16.0 & 18.4 \\ 16.0 & 29.3 & 33.4 \end{pmatrix} \tag{1.34}$$